

LED Behavior Specification

BLE Pairing Visual Feedback Guide

Version 1.0 — March 2026

For Hardware, Firmware & App Teams

State	Color	Pattern
01. DISCOVERABLE	Brand Blue	Slow breathing pulse
02. FOUND	Brand Blue	Faster breathing pulse
03. PAIRING IN PROGRESS	Brand Purple	Rapid double-flash
04. AWAITING CODE	Brand Amber	Steady solid glow
05. CODE VERIFICATION	Brand Magenta	Fast chase sweep
06. PAIRING SUCCESS	Brand Blue → White	Triple celebration flash → solid
07. WIFI CONNECTING	Brand Blue	Slow wave ripple

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08. SETUP COMPLETE	Brand Gradient → White	Brand gradient sweep → solid white
09. PAIRING FAILED	Safety Red	Three slow red pulses → discoverable
10. E-STOP ACTIVE	Safety Red + Yellow	Rapid red/yellow alternating flash

STATE 01

DISCOVERABLE

Brand Blue

#3B5EE8

WHEN

Robot is powered on and not yet paired.

App is on BLE Scan screen.

APP SCREEN

02a — Find Your Robot

LED PATTERN

Slow breathing pulse

Brightness fades smoothly in and out.

3-second cycle (1.5s in, 1.5s out)

Brightness: 20% → 80% → 20%

PURPOSE

"I'm here, I'm ready to be found."

The slow pulse is calm and non-urgent — the robot is waiting patiently.

SCRIPT TRIGGER

```
Default state when BLE advertising is active and no device is connected.
```

STATE 02

FOUND

Brand Blue

#3B5EE8

WHEN

App has detected the robot via BLE scan.

Robot appears in the discovered list.

APP SCREEN

02a — Find Your Robot (robot card visible)

LED PATTERN

Faster breathing pulse

Same fade pattern, quicker rhythm.

1.5-second cycle (0.75s in, 0.75s out)

Brightness: 30% → 100% → 30%

PURPOSE

"Yes, I see you too."

Helps confirm the right robot if multiple are nearby.

SCRIPT TRIGGER

```
BLE scan response sent to a device.
```

STATE 03

PAIRING IN PROGRESS

Brand Purple

#8B45B5

WHEN

User tapped "Pair This Robot".

BLE connection is being established.

APP SCREEN

02b — Pairing

LED PATTERN

Rapid double-flash

Two quick flashes then a pause, repeating.

Flash 100ms on, 100ms off, 100ms on, 600ms off

1-second total cycle. Sharp on/off, not faded.

PURPOSE

"I'm processing the connection."

Purple differentiates from discovery blue.

Double-flash is distinct and unmistakable.

SCRIPT TRIGGER

```
BLE connection established (GATT connection callback).
```

STATE 04

AWAITING CODE

Brand Amber

#D4A012

WHEN

BLE connected. Robot displays its 6-digit code.

App shows the code entry screen.

APP SCREEN

02b — Pairing (code entry fields)

LED PATTERN

Steady solid glow

No animation. Constant brightness.

70% steady. No change until code entered.

2-minute timeout.

PURPOSE

"I'm waiting for you to enter the code."

Amber is warm and patient.

Steady means "your turn."

SCRIPT TRIGGER

```
Pairing code generated and sent via BLE characteristic.
```

STATE 05

CODE VERIFICATION

Brand Magenta

#D63384

WHEN

User entered the code. Robot is verifying.

APP SCREEN

02b — Pairing (loading after code)

LED PATTERN

Fast chase sweep

Lights sweep left to right across strip.

Full sweep in 500ms, repeating.

Leading LED 100%, trailing fade to 20%.

PURPOSE

"I'm checking..."

Chase pattern conveys computation.

Brief — lasts only 1-2 seconds.

SCRIPT TRIGGER

```
Code received via BLE write. Verification called.
```

STATE 06

PAIRING SUCCESS



Brand Blue → White

#3B5EE8 → #E8EAED

WHEN

Code verified. Secure BLE link established.

APP SCREEN

02c — Wi-Fi Setup (success banner)

LED PATTERN

Triple celebration flash → solid

Three quick flashes (blue), transition to solid white.

Flash 150ms on / 100ms off × 3

Then 500ms transition to solid white at 60%.

PURPOSE

"Success!"

Triple-flash is a clear celebration.

White means "paired and ready."

SCRIPT TRIGGER

```
Code verification returns True. State → PAIRED.
```

STATE 07

WIFI CONNECTING

Brand Blue

#3B5EE8

WHEN

Robot connecting to user's home Wi-Fi.

APP SCREEN

02c — Wi-Fi Setup (connecting)

LED PATTERN

Slow wave ripple

Brightness ripples across strip like a wave.

Wave traverses full strip in 2 seconds.

Peak 80%, trough 20%.

PURPOSE

"I'm connecting to your network."

Wave suggests data flowing.

Distinct from breathing pulse.

SCRIPT TRIGGER

```
Wi-Fi credentials received. Connection attempt started.
```

STATE 08

SETUP COMPLETE



Brand Gradient → White

Blue→Purple→Magenta→Red→Pink

WHEN

Wi-Fi connected (or skipped). Fully paired.

APP SCREEN

02d — Setup Complete

LED PATTERN

Brand gradient sweep → solid white

Full brand spectrum sweeps across strip.

2-second sweep (one pass left to right).

Settles to solid white at 50%.

PURPOSE

The grand finale. Brand gradient sweep mirrors the logo. "I'm fully ready to work." Solid white = idle/ready going forward.

SCRIPT TRIGGER

```
Wi-Fi confirmed or skip received. State → READY.
```

STATE 09

PAIRING FAILED

Safety Red

#FF3B30

WHEN

Code wrong, BLE dropped, or 2-min timeout.

APP SCREEN

Error dialog / return to BLE Scan

LED PATTERN

Three slow red pulses → discoverable

Pulse 500ms on / 300ms off × 3 (2.4s total).

1s pause. Then resume State 1 (discoverable).

100% brightness — unmissable.

PURPOSE

"Something went wrong — let's try again."

Returns to discoverable automatically.

No power cycle needed.

SCRIPT TRIGGER

```
Verification False, BLE disconnect, or timeout.
```

STATE 10

E-STOP ACTIVE

Safety Red + Yellow

#FF3B30 / #FFD600

WHEN

Emergency stop triggered from any source.

APP SCREEN

Any screen — E-Stop activated

LED PATTERN

Rapid red/yellow alternating flash

200ms red / 200ms yellow, continuous.

100% brightness. Visible from 20+ meters.

Overrides ALL other states. HIGHEST PRIORITY.

PURPOSE

SAFETY CRITICAL.

Matches industrial E-Stop convention.

Impossible to ignore or mistake.

SCRIPT TRIGGER

```
E-Stop via BLE, WiFi, MQTT, or physical button.
```

STATE MACHINE

LED State Transitions

```
POWER ON → [1. DISCOVERABLE] → BLE scan → [2. FOUND] → Pair tap → [3. PAIRING]
→ Connected → [4. AWAITING CODE] → Code entered → [5. VERIFICATION]
→ Success → [6. PAIRING SUCCESS] → [7. WIFI CONNECTING] → [8. SETUP COMPLETE]
→ Failure → [9. PAIRING FAILED] → back to [1. DISCOVERABLE]

[10. E-STOP] can be triggered from ANY state. HIGHEST PRIORITY.
```

HARDWARE REQUIREMENTS

Requirement	Specification
LED Type	WS2812B / SK6812 (addressable RGB)
LED Count	30-60 LEDs across front accent bar
Controller	Jetson GPIO (PWM) or SPI via NeoPixel library
Power	5V, ~1A max (60 LEDs at full white)
Viewing Distance	Patterns distinguishable from 5m in daylight
Weatherproofing	IP65 LED strip + sealed housing